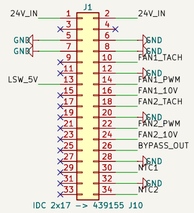
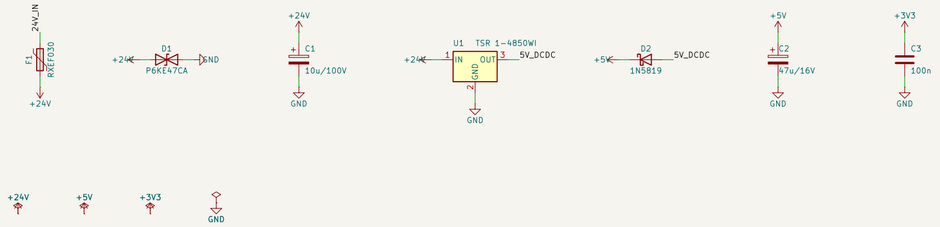
J1 — 34-pin ribbon to power board 439155 (J10)



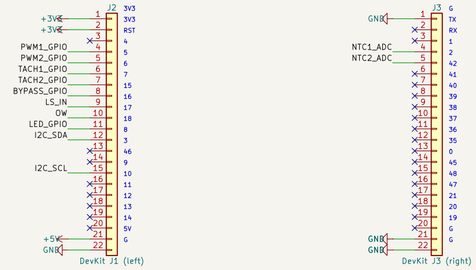
Measured 16.10.2026: 1/2 = 24 V (29 V no load), 3/4 = +5 V (not used).
FAN1 (EXTRACT): 10 tacho, 12 motor GND, 14 PWM input (0 V = stop), 16 +10 V.
FAN2 (SUPPLY): 18, 20, 22, 24 in the same order. Motor GNDs isolated on 439155 -> tied to GND.
13 = LS (ISP814, 5 V pull-up 2k2 on 439155), 26 = bypass relay drive (BC337 base).
28/30 = NTC TEMP1 (extract air), 32/34 = NTC TEMP2 (outdoor intake): 28/32 to GND, 30/34 to the ADC.
15/17/19/21 = original SHT1x humidity sensor — not used.

Power: 24 V -> PTC -> TVS -> TRACO TSR 1-4850WI 5 V -> Schottky -> DevKit 5V pin



F1 PTC RXEF030: 72 V, hold 0.30 A / trip 0.60 A at 20 C (hold ~0.20 A at 60 C), pitch 5.1 mm; alternative
0ZRE0033FF1A (240 V, 0.33/0.74 A) fits the same holes. Both:
a SLOW, temperature dependent limiter for a shorted D1/U1, not a fast fuse. Unit draw < 0.2 A from 24 V.
D1 P6KE47CA (bidirectional, DO-15): standoff 40.2 V > 29 V, clamp 64.8 V < 72 V TSR max (as SMAJ40A in D1).
U1 TRACO TSR 1-4850WI 9-72 V -> 5 V / 1 A, SIP-3 (1 = IN, 2 = GND, 3 = OUT) + 2 case pins, body 12.1x8.6x17.5 mm;
no tracks under the converter (TRACO drawing rev. 2026-07-02, review E-01).
D2 1N5819: the DevKit must not get 5 V from both the 5V pin and its USB at once (Espressif: 'mutually
exclusive') — the diode keeps the USB 5 V out of the TSR. DevKit 5V pin ~4.6 V is enough for its LDO.
+3V3 = the DevKit's own 3.3 V regulator (3V3 pins): pull-ups, NTC dividers, I2C / 1-Wire sensors.

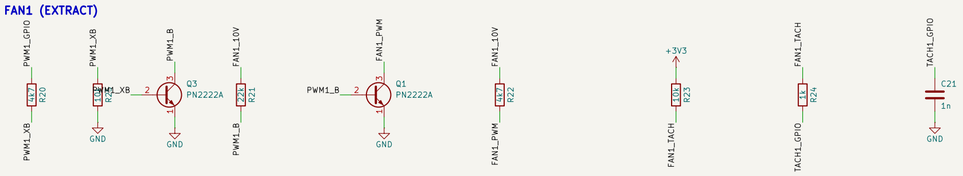
ESP32-S3-DevKitC-1-N8R8 (or N8/N16R8) in 2x 1x22 sockets 8.5 mm, rows 22.86 mm apart



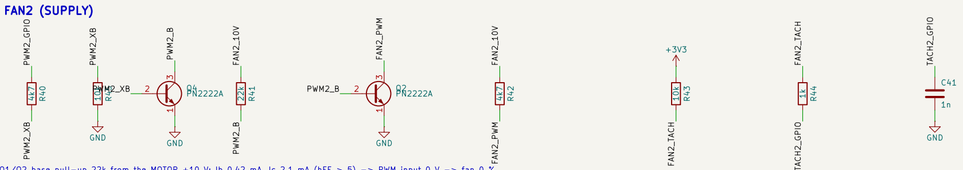
Pin names = GPIO numbers. Unused: strapping 0/3/45/46, USB 19/20, UART0 43/44 (log).
RGB LED 38 (v1.1) / 48 (v1.0), 35-37 (octal PSRAM on R8 variants -> N8R8 is fine), RST.
NB is EOL at Espressif (review E-06): order ESP32-S3-DevKitC-1-N8R8 v1.1.

Fans: stop unless a live GPIO says run (stop bias from the motor +10 V), tacho input

FAN1 (EXTRACT)

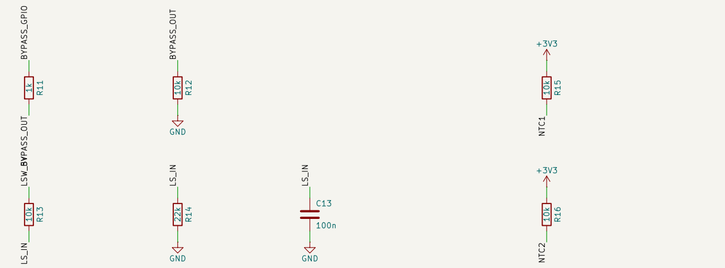


FAN2 (SUPPLY)



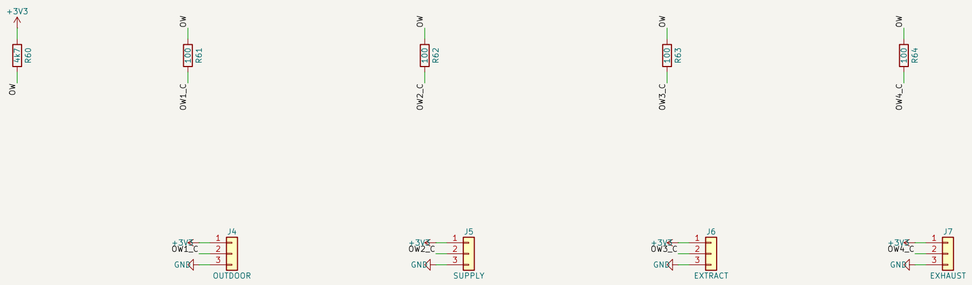
Q1/Q2 base pull-up 22k from the MOTOR +10 V: Ib 0.42 mA, Ic 2.1 mA (hFE > 5) -> PWM input 0 V -> fan 0 %.
Q3/Q4 pull that base low only while the GPIO drives HIGH (4k7, Ib 0.48 mA) -> 4k7 lifts the PWM -> fan runs.
Firmware: NON-inverted PWM (GPIO high = run). Reset / high-Z, DevKit unplugged, 3V3 lost or shorted,
+5V lost (F1, U1, D2) or ESP unpowered -> Q3/Q4 off (10k to GND) -> fans 0 % as long as the motors have 10 V.
No path depends on an unpowered GPIO (preorder review P2).
A HUNG program keeps its last PWM: the firmware must keep the ESP32 task watchdog enabled.
Tacho: open collector, 1 pulse/rev, 10k pull-up to 3.3 V, 1k + 1n low-pass to the GPIO (PCNT).

Bypass, light switch input, two NTCs on their own ADC inputs, status LED



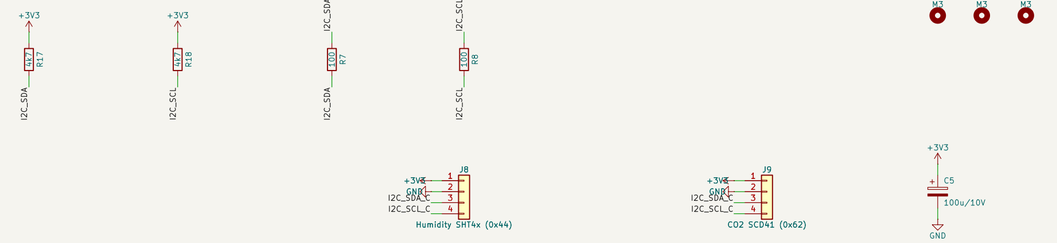
BYPASS_OUT -> BC337 + relay RL1 on 439155 (230 V damper actuator). GPIO15 is high-Z at reset (no pull on the
DevKit): R12 10k holds the BC337 base low -> relay off in reset, bootloader and without the DevKit (E-03).
LS: ISP814 pulls 100 Hz pulses low while the light switch is live: 10k/22k -> 3.4 V max.
NTC 5k @ 25 C (B ~3950) between J1 28/30 (TEMP1) and 32/34 (TEMP2), 10k pull-up to 3.3 V,
each on its own ADC1 channel (GPIO1 / GPIO2): no NTC multiplexing, no CH340 back-feed (alpha issue).
Open NTC -> 3.3 V, short -> 0 V: both rejected by the firmware.

4x DS18B20 on one 1-Wire bus (GPIO17), 4k7 pull-up, 100R per connector



JST-XH 3-pin: 1 = 3V3, 2 = DATA, 3 = GND (same as rev D). Series 100R per connector LIMIT the current into the
ESP32 pin clamps; they do not set a safe clamp voltage. No dedicated TVS (review E-04: open decision, depends on
the cable lengths — all sensor cables stay inside the unit).

I2C (GPIO8 SDA, GPIO9 SCL): SHT4x humidity (J8), SCD41 CO2 (J9)



JST-XH 4-pin: 1 = 3V3, 2 = GND, 3 = SDA, 4 = SCL (same as rev C/D). R7/R8 100R in series to the GPIOs.
C5 next to J9 on the PCB (SCD41 ~200 mA peaks from the DevKit 3.3 V regulator).

Silkscreen: VA MVHR control board rev E (no manufacturer name)
Through-hole only, hand assembly. Interface proven by the alpha prototype 16.10.2026
Connects to J10 (34-pin ribbon) of power board 439155, replaces control board 439150

Sheet: /
File: mvhr_ctrl.kicad_sch

Title: MVHR control board — ESP32-S3-DevKitC-1, through-hole

Size: A2	Date:	Rev: E
KiCad E.D.A. 10.0.7		Id: 1/1